

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01

Parent: ^{35}Ar : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=1.7755$ s 14; $Q(\varepsilon)=5966.2$ 7; $\%\varepsilon+\%\beta^+$ decay=100.0

^{35}Ar - J^π , $T_{1/2}$: From Adopted Levels of ^{35}Ar .

^{35}Ar - $Q(\varepsilon)$: From 2021Wa16.

1980Wi13: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 11.0-MeV protons from the ONR-CIT tandem accelerator impinging on LiCl and PbCl_2 wrapped in 25- μm Ta foil. γ rays were detected using a 100- cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 20 γ -ray peaks. Deduced decay branching ratios for 9 levels, and $\log ft$. Compared with shell-model calculations.

1984Ad01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 13-MeV protons from the University of Washington FN tandem accelerator impinging on a thick KCl target. γ rays were detected using a 28% $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 8 γ -ray peaks. Deduced decay branching ratios for 6 levels, and $\log ft$. Compared with shell-model calculations.

1985Da04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 9-MeV protons from the Van de Graaff accelerator at the University of Pittsburgh impinging on a NaCl target. γ rays were detected using a $\text{Ge}(\text{Li})$ detector with NaI anti-Compton shielding. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 3 levels.

1979Ha04: ^{35}Ar was produced via the $\text{V}(p,p11n)$ spallation reaction with 600-MeV protons from the CERN synchrocyclotron bombarding a 38- g/cm^2 vanadium target. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for the 1219 γ . Deduced the decay branching ratio for the 1219-keV level.

1977Az01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-13-MeV protons from the McGill synchrocyclotron impinging on CCl_4 and LiCl targets and 8-MeV protons from the University of Montreal impinging a thick LiCl pellet. Positrons were detected using a plastic (NE-102) coupled to an RCA 6342A photomultiplier tube. γ rays were detected using a 18% $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 3 γ -ray peaks. Deduced decay branching ratios for 3 levels and parent ^{35}Ar $T_{1/2}$. 1978Az01: $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction threshold.

1971Ge04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the Chalk River tandem accelerator impinging on PbCl_2 targets. γ rays were detected using two coaxial $\text{Ge}(\text{Li})$ detectors. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 4 levels, decay branching ratios, $\log ft$, and parent ^{35}Ar $T_{1/2}$.

1971De05: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction on KCl targets. γ rays were detected using a 25 cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ . Deduced 6 levels, decay branching ratios, and $\log ft$.

1969Wi18,1969Fr09: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with a proton beam from the Harwell tandem accelerator impinging on 3.75- mg/cm^2 KCl deposited on 0.13-mm Au backings and 160-mmHg gas CCl_4 targets. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 2 γ -ray peaks. Deduced 2 levels, decay branching ratios, (p,n) reaction threshold and β end-point energy, $\log ft$, and parent ^{35}Ar $T_{1/2}$.

1960Wa04: ^{35}Ar was produced at Crocker Laboratory and the Lawrence Radiation Laboratory. β spectra were measured with a uniform-held 180°-deflection single-focusing spectrometer. Deduced Coulomb-energy differences between mirror ground-state transitions.

1956Ki29: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the BNL 60-inch cyclotron impinging on a liquid CCl_4 target. β particles were detected using a conventional thin-lens, iron-free, magnetic spectrometer. γ rays were detected using a $\text{NaI}(\text{Ti})$ crystal scintillation counter. Measured E_β and total I_β for 2 positron decays to ^{35}Cl excited states, E_γ and I_γ for 2 γ -ray peaks. Deduced 2 levels and decay branching ratios.

^{35}Ar β asymmetry parameter measurements: 1993Co07, 1991Na05, 1991Mi18, 1988Ga02, 1965Ca04.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	$3/2^+$	stable
1219.2 2	$1/2^+$	
1763.1 2	$5/2^+$	
2645.1 10	$3/2^+$	
2693.8 2	$3/2^+$	
3002.3 3	$5/2^+$	
3918.4 5	$3/2^+$	
3967.7 4	$1/2^+$	
4624.5? 5	$(3/2, 5/2^+)$	

† From a least-squares fit to γ -ray energies.

‡ From the Adopted Levels.

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ε, β^+ radiations						Comments
E(decay)	E(level)	$I\beta^+ \ddagger$	$I\varepsilon \ddagger$	Log ft	$I(\varepsilon+\beta^+) \ddagger\ddagger$	
(1341.7 [#] 13)	4624.5?	$<8.0 \times 10^{-4}$	<0.0035	>3.8	<0.0043	$I(\varepsilon+\beta^+)$: weighted average of 1.223 46 (1969Wi18,1969Fr09), 1.22 20 (1971Ge04), 1.55 15 (1977Az01), 1.34 8 (1979Ha04), 1.23 7 (1984Ad01), 1.228 30 (1980Wi13). $I(\varepsilon+\beta^+)$: deduced from $100 - \Sigma I(\varepsilon+\beta^+)$ (excited states), where $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.79 6. Level scheme fed by β^+ decay may be complete considering the highest observed level is at 4.6 MeV. Observed $\Sigma I(\varepsilon) \approx 0.08$. Evaluators assume $\Sigma I(\varepsilon) \leq 0.08$ for all unobserved ε feedings to high-lying levels and include this limit in the uncertainty for $\Sigma I(\varepsilon+\beta^+)$ (excited states) and $I(\varepsilon+\beta^+)$ (gs).
(1998.5 13)	3967.7	0.015 11	0.0016 10	4.5	0.017 11	
(2047.8 13)	3918.4	0.0097 34	8.4×10^{-4} 27	4.8	0.0105 34	
(2963.9 13)	3002.3	0.090 15	0.0011 2	5.0	0.091 15	
(3272.4 13)	2693.8	0.166 10	0.00127 8	5.0	0.167 10	
(3321.1 16)	2645.1	<0.00149	$<1.1 \times 10^{-5}$	>7.1	<0.0015	
(4203.1 13)	1763.1	0.259 16	7.1×10^{-4} 5	5.5	0.260 16	
(4747.0 13)	1219.2	1.24 6	0.0021 1	5.10 2	1.24 6	
(5966.2 16)	0	98.14 10	0.0731 8	3.760 1	98.21 10	

[†] Deduced from γ -ray intensity balance, relative to $I_\gamma(1219\gamma)=100$, normalized to absolute $\%I_\gamma(1219\gamma)=1.26$ 5.

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

 $\gamma(^{35}\text{Cl})$

I_γ normalization: absolute $\%I_\gamma(1219\gamma)=1.26$ 5 deduced from weighted average of 1.223 46 (1969Wi18,1969Fr09), 1.22 20 (1971Ge04), 1.55 15 (1977Az01), 1.34 8 (1979Ha04), 1.23 7 (1984Ad01), 1.228 30 (1980Wi13).

$E_\gamma \dagger$	$I_\gamma \ddagger @$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
916 ^{&}		3918.4	3/2 ⁺	3002.3	5/2 ⁺	The intensity could not be determined due to being obscured by another peak (1980Wi13).
930.7 2	1.57 20	2693.8	3/2 ⁺	1763.1	5/2 ⁺	$\%I_\gamma=0.0198 +34-32$ I_γ : weighted average of 1.13 26 (1980Wi13) and 1.66 12 (1984Ad01).
965 ^{&}	$<0.87^\#$	3967.7	1/2 ⁺	3002.3	5/2 ⁺	$\%I_\gamma<0.0110$
1219.3 2	100.0 15	1219.2	1/2 ⁺	0	3/2 ⁺	$\%I_\gamma=1.26$ 6 I_γ : others: 100.0 30 from 1984Ad01, 100 10 from 1977Az01, 100 from 1969Wi18, 1971Ge04, 1979Ha04, and 1985Da04.
1225 ^{&}		3918.4	3/2 ⁺	2693.8	3/2 ⁺	The intensity could not be determined due to being obscured by another peak (1980Wi13).
1239 ^{&}	$<0.30^\#$	3002.3	5/2 ⁺	1763.1	5/2 ⁺	$\%I_\gamma<0.0038$
1274 ^{&}	$<0.42^\#$	3967.7	1/2 ⁺	2693.8	3/2 ⁺	$\%I_\gamma<0.0053$
1474.8 3	1.02 11	2693.8	3/2 ⁺	1219.2	1/2 ⁺	$\%I_\gamma=0.0129$ 19 I_γ : weighted average of 0.83 13 (1980Wi13) and 1.08 7 (1984Ad01).
1763.0 2	22.7 4	1763.1	5/2 ⁺	0	3/2 ⁺	$\%I_\gamma=0.286$ 13 I_γ : weighted average of 23.4 13 (1969Wi18), 20.0 27 (1971Ge04), 22 6 (1977Az01), 23.08 37 (1980Wi13), 21.85 67 (1984Ad01), and 21.7 9 (1985Da04).

Continued on next page (footnotes at end of table)

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) [1980Wi13](#),[1984Ad01](#) (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^{\ddagger@}$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
1783&	<0.39#	3002.3	5/2 ⁺	1219.2	1/2 ⁺	%I γ <0.0049
1931&	<0.21#	4624.5?	(3/2,5/2 ⁺)	2693.8	3/2 ⁺	%I γ <0.0027
2155.1 15	0.15 6	3918.4	3/2 ⁺	1763.1	5/2 ⁺	%I γ =0.0019 +9-8 I γ : 0.15 6 (1984Ad01). Other: <0.22 (1980Wi13).
2204&	<0.27#	3967.7	1/2 ⁺	1763.1	5/2 ⁺	%I γ <0.0034
2645&	<0.12#	2645.1	3/2 ⁺	0	3/2 ⁺	%I γ <0.00151
2693.7 2	10.96 18	2693.8	3/2 ⁺	0	3/2 ⁺	%I γ =0.138 6 I γ : weighted average of 10.2 15 (1971Ge04), 10.96 18 (1980Wi13) and 11.02 34 (1984Ad01).
2699&	<0.33#	3918.4	3/2 ⁺	1219.2	1/2 ⁺	%I γ <0.0042
2748.5 6	0.477 34	3967.7	1/2 ⁺	1219.2	1/2 ⁺	%I γ =0.0060 5 I γ : weighted average of 0.48 6 (1980Wi13) and 0.476 34 (1984Ad01).
2861&	<0.099#	4624.5?	(3/2,5/2 ⁺)	1763.1	5/2 ⁺	%I γ <0.00125
3002.1 4	7.29 14	3002.3	5/2 ⁺	0	3/2 ⁺	%I γ =0.092 4 I γ : weighted average of 9.0 15 (1971Ge04), 7.24 14 (1980Wi13), 7.45 23 (1984Ad01) and 6.9 5 (1985Da04).
3405&		4624.5?	(3/2,5/2 ⁺)	1219.2	1/2 ⁺	I γ : could not be determined due to being obscured by another peak (1980Wi13).
3918	0.52 3	3918.4	3/2 ⁺	0	3/2 ⁺	%I γ =0.0066 5
3968	0.106 25	3967.7	1/2 ⁺	0	3/2 ⁺	%I γ =0.0013 4
4624&	<0.035#	4624.5?	(3/2,5/2 ⁺)	0	3/2 ⁺	%I γ <4.4×10 ⁻⁴

[†] From [1984Ad01](#), unless otherwise noted.

[‡] from [1980Wi13](#), unless otherwise noted.

[1980Wi13](#) gives upper limits of I γ at 98% confidence level. These peaks are not observed in the spectra of [1980Wi13](#) or in [1984Ad01](#).

@ For absolute intensity per 100 decays, multiply by 0.0126 5.

& Placement of transition in the level scheme is uncertain.

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Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
 — $I_\gamma < 10\% \times I_\gamma^{\max}$
 — $I_\gamma > 10\% \times I_\gamma^{\max}$
 - - - γ Decay (Uncertain)

Decay Scheme

Intensities: I_γ per 100 parent decays